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Palaeogeographic implications of differential inclination shallowing in permo-carboniferous sediments from the donets basin, Ukraine

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ABSTRACT

We present new palaeomagnetic data from Upper Carboniferous and Lower Permian grey and red sediments from the Donets Basin, Ukraine, part of the Palaeozoic East European Platform. Detailed demagnetization of these units reveals two ancient components of magnetization: component "B", which is carried by magnetite and pigmentary haematite, and a high unblocking temperature component "C", present only in the red beds, carried by detrital haematite. The "B" and "C" components both pass fold tests indicating a primary or a near primary-origin for magnetizations. The "C" component, however, yields palaeolatiudes that are consistently lower (by up to ~12° of latitude or ~1330 km) than those derived from the "B" component, and we argue that this is due to significant inclination-shallowing of the "C"-component. A comparison with European reference palaeomagnetic data reveals that the reference data also span a large spread of palaeolatitudes for this time, and we argue that unrecognized shallowing may have crept into the reference data when based on sedimentary units. A more rigorous approach to selecting reference palaeomagnetic data may well be key to resolving palaeogeographic controversies at this time.

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1. Introduction

Continental red beds make up a significant proportion of the Palaeozoic sedimentary record and are often relatively strongly magnetized. Therefore they are commonly used for palaeomagnetic studies, and results from red beds dominate the palaeomagnetic database for several Palaeozoic continents (e.g. Van der Voo, 1993; McElhinny and Lock, 1996). Despite this reliance on results from red bed sequences aspects of their remanence acquisition remain unclear (e.g. Tan and Kodama, 2002), and, hence, our confidence in the reliability of these results needs to be tempered with caution. Nevertheless, magnetostratigraphic, palaeomagnetic, and rock magnetic studies on red bed sequences from a broad variety of locations have suggested that a primary magnetic reversal stratigraphy may be preserved, because magnetic reversals within these sequences can be correlated over large distances (e.g. Khramov et al., 1974; Van den Ende, 1977; Steiner, 1983; Tauxe and Badgley, 1984; Rodionov and Osipova, 1985; Molina-Garza et al., 1991; Maillol and Evans, 1992; Channell et al., 1993; Kent et al., 1995; Kruiver et al., 2000; Abdul Aziz et al., 2000, Abdul Aziz et al., 2003; Rodionov et al., 2003; Abdul Aziz and Langereis, 2004). These results suggest that many red bed sequences can be used for geodynamic interpretation and/or regional stratigraphic correlation, an asset because red beds typically contain few fossils.

Constraining the timing of the natural remanent magnetization (NRM) in red beds is, however, not always straightforward: they often contain varying mixtures of detrital haematite, in the form of "specularite", and pigmentary haematite, which is formed during diagenetic processes (e.g. Collinson, 1965; Turner et al., 1999). The NRM can, therefore, incorporate components of magnetization that date not only from the time the rock was formed, but also from components that have been added during the growth of pigmentary haematite during diagenesis. Hence, the ratio of the detrital remanent magnetization (DRM) and chemical remanent magnetization (CRM) may vary between sample collections from different areas and even within a single collection. An example of regionally varying NRMs in a single formation is that of the Triassic Moenkopi Formation: in Colorado (USA) Larson et al. (1982) argued for a dominant CRM character of the NRM while in New Mexico (USA) Molina-Garza et al. (1991) could show a DRM nature of the NRM by means of an intraformational micro-conglomerate test. Moreover, magnetite or partially maghemitized magnetite can be part of the NRM and goethite may also be present in certain red beds (e.g. Kruiver et al., 2003; Grygar et al., 2003; Abdul Aziz and Langereis, 2004). This

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varying nature of the acquisition process of red bed NRMs has been widely referred to as the "red bed controversy" (e.g. Butler, 1992); it has prompted an evaluation of the NRM origin on a case by case basis. Recent examples of such evaluations are given in Kruiver et al. (2002) detailing the NRM behaviour of Late Miocene red bed sediments in Southern Spain and in Rodionov et al. (2003) detailing the NRM behaviour of Middle Ordovician red beds on the Siberian platform.

The time period over which the NRM of red beds is acquired may therefore be very long (e.g. Khramov et al., 1974; Larson et al., 1982). In the Ammonitico Rosso Formation (pelagic limestone) Channell et al. (1982) identified two NRM components, one residing in magnetite (DRM) and another residing in pigmentary hematite (CRM) that recorded the same geomagnetic reversal at different stratigraphic levels. Dual polarities have also been found in single red bed specimens (Khramov, 1986; Iosifidi et al., 1999), indicating that remanence acquisition may have spanned a considerable amount of time, at least a substantial part of two subsequent polarity zones.

When dealing with a detrital remanent magnetization, an additional complication arises from the fact that the sediments may also be prone to inclination-shallowing, where large detrital grains are subject to mechanical rotation during deposition and later compaction (e.g. Tan et al., 2002). On the other hand, if inclination shallowing can be proven, it would be a strong indication for the DRM nature of that NRM component, thus providing a powerful constraint on its age. When shallowed inclinations go unnoticed one ends up with incorrect paleolatitudinal inferences. In laboratory redeposition experiments large inclination shallowing (>15°) is observed for hematite-bearing sediments (Lovlie and Torsvik, 1984; Tauxe and Kent, 1984; Tan et al., 2002). Also shallowed inclinations are demonstrated in many studies of natural red bed sequences (e.g. Tauxe and Kent, 1984; Stamatakos et al., 1995; Garces et al., 1996; Rösler and Appel, 1998; Gautam and Fujiwara, 2000; Ojha et al., 2000; Gilder et al., 2001). A recent example is that of Tan et al. (2003) who showed for Cretaceous red beds from the Kapusaliang Group (Tarim Basin, China) that shallow inclinations are caused by depositional processes rather than tectonic displacement of about 1000 km Northward.

To correctly interpret the history of NRM acquisition of the rock, and hence the validity of using the NRM components for stratigraphic correlation or for tectonic applications, the acquisition mechanism and age of each NRM component must be determined. This aspect is the most difficult part of any palaeomagnetic study, but is vital for verifying whether a magnetic component is of the same age as its host rock. One approach to this problem is to try to assess the fidelity of the record from sequences where red beds of different grain size populations are present (e.g. Mac Niocaill, 2000), or are intercalated with other lithologies. In such cases the diagenetic history of the reddened and unreddened sediments will be different, and differences in remanence components might enable a better delineation of the mode of origin of the NRM.

We present the results of a detailed palaeomagnetic study of intercalated, variegated, sandstones and clays, from the Upper Carboniferous and Lower Permian of the Donets Basin, Ukraine (Fig. 1). We chose this setting for three main reasons: (1) because the sequence contains a range of lithologies, and hence the possibility of evaluating the fidelity of the palaeomagnetic record in the differing lithologies; (2) previously published data from the Kartamysh Formation of the Donets Basin (Khramov, 1963; Khramov et al., 1974; Iosifidi and Khramov, 2002) had indicated that the rocks carried stable components of magnetization, though only the most recent work had employed stepwise progressive demagnetization techniques (Iosifidi and Khramov, 2002); and (3) the region underwent folding originally thought to be due to compression in the late Early Permian but recently argued to be caused by salt tectonics in a transtensional setting (Stephenson et al., 2006). Two post-Paleozoic compressional phases have been proposed (Stovba and Stephenson, 1999; Saintot et al., 2003), one of late Triassic–early Jurassic age and one of late Cretaceous–early Paleogene age, that inverted the basin and ultimately led to the exposure of the strata. Nonetheless, a positive fold test would still meaningfully bracket the age of corresponding NRM components. A recent study by Meijers et al in press of the same area primarily focuses on geodynamic implications of the paleopoles after unflattening using the elongation/inclination (E/I) method (Tauxe and Kent , 2004). The present study involves an analysis from different locations (only one site is the same in both independent studies) and focuses on differences in demagnetization behaviour as function of lithology.

Finally, we also compare our data with the previously available data for the East European Platform (EEP) for Permo-Carboniferous times. The amount of relevant data that meet modern reliability criteria is surprising low, with only 8 out 50 available data in the global palaeomagnetic data passing 4 or more of the Van der Voo (1988) reliability criteria. Furthermore, of the poles that do appear to be of moderate reliability the bulk were derived from igneous rocks in the Northwestern EEP, and may not have sufficiently averaged out secular variation. Hence, these poles should be regarded as virtual geomagnetic poles (VGPs) rather than as true palaeomagnetic poles.

2. Geological setting

The sedimentary series sampled forms the upper part of a thick sequence of terrigenous and carbonate sediments, with interbedded coal measures in the Donets basin, Ukraine. This basin represents the uplifted, east–south–east termination of the Dniepr–Donets aulacogen, which can be traced to the west across Ukraine to the edge of the East European platform (Fig. 1b). The Dniepr–Donets aulacogen, a rejuvenated Riphean rift structure located between the Azov and Voronezh Precambrian massifs, forms part of a back-arc extensional system that was active during Variscan times, with the initial stages of activity being in the early Devonian (Ziegler, 1989; Zonenshain et al., 1990; Khain and Seslavinsky, 1991). In the late Visean the geodynamic history of the main part of the aulacogen and the Donets basin diverged sharply (Khain, 1977), with the aulacogen experiencing a platform regime at that time whereas the Donets basin underwent a phase of rapid subsidence. During the Carboniferous thick, paralic, molasse-type, coal-bearing formations accumulated, with a total thickness exceeding 20 km. At the end of the Carboniferous, the central part of the Donets basin started to be uplifted. At its northern rim the foreland type sedimentation ceased only in the Early Permian (Khain, 1977; Khain and Seslavinsky, 1991).

In the sampling area more than 3.5 km of Upper Carboniferous and Lower Permian sediments were accumulated, with an increasing proportion of red beds starting to appear in the Late Carboniferous. Uplift and deformation of the region, associated with the convergence of Hercynian massifs with the East European Platform, commenced in the Artinskian stage (late Early Permian), with the most intense folding occurring during the latest Early Permian (Kungurian stage, cf. Khain and Seslavinsky, 1991) although more recent work (Stovba and Stephenson, 1999; Saintot et al., 2003; Stephenson et al., 2006) argue for younger ages of the folding. As a result, folded Carboniferous and Lower Permian strata outcrop in several locations within the Donets Basin, forming synclinal structures. Our sampling area is located on the northern flank of this large synclinorial structure (Fig. 1).

3. Sampling

We examined specimens from collections of the Upper Carboniferous and Lower Permian sedimentary rocks of the western Donets Basin (Bakhmut Basin) that were sampled in the early 1960 s (Khramov, 1963). The ages of sampled formations are constrained by a rich shallow marine fauna of brachiopods, pelecipodes and cephalopods, which occur in the thin carbonate beds, and by the presence of plant remnants of cordaites and paracalamites in the terrigenous members (Lapkin, 1961; Stratigrafiya USSR, 1970). The Late Carboniferous Avilov and

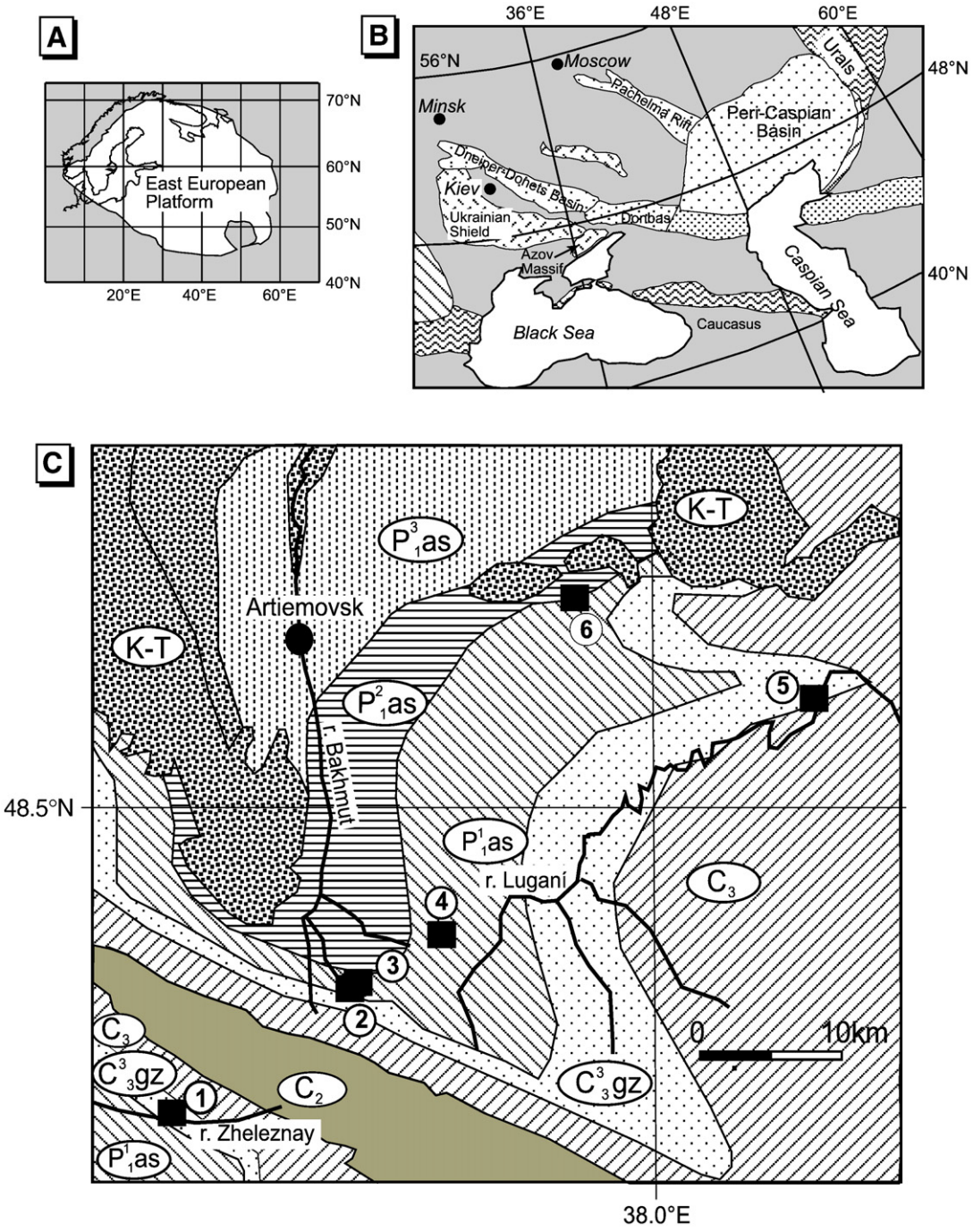


Fig. 1. Location on the East European Platform (A) and schematic map of main geological structures in the region (B, after Yegorova et al., 2004) and geological map of the Bakhmut river basin area, Donbass region of Ukraine with the sampled sections (C). K–T = Cretaceous–Tertiary cover sediments; P^1_1as, P^2_1as, and P^3_1as = Early Permian Asellian stages; C_2, C_3 Late Carboniferous, C^3_3gz = Late Carboniferous Gzelian stage. Section numbers: 1 = river Zheleznaya, 2 = village Zaeitsevo, 3 = village Sukhoi Yar, 4 = village Dolomit, 5 = village Kalinovo, 6 = village Pilipchatyi. B edding attitudes (dip azimuth/dip) are 318–337°/10–68° (village Pilipchatyi section), 017–032°/32–56° (village Sukhoi Yar section), 180–196°/45–82° (village Kalinivo section), 028°/11° (village Zaetsevo section), 165°/10° (village Dolomit section), and 206°/70° (river Zheleznaya section).

Araucarite Formations (Gzelian age, Nesterenko, 1956; Stratigrafiya USSR, 1970; Movshovich and Redichkin, 1979; cf. Fig. 2) were selected from outcrops along the river Lugan valley near the villages of Kalinovo and Zeleznaya. We collected 9 samples (18 specimens) of grey beds from 5 levels and 1 sample (2 specimens) of red beds from 1 level (Fig. 2) from the Avilov Formation, and 17 samples (29 specimens) of grey beds from 7 levels and 9 samples (18 specimens) of red beds from 8 levels from the Araucarite Formation. Lower Permian (Asselian stage) rocks were collected from outcrops near the settlements of Dolomit, Sukhoi Yar, Pilipchatyi and Zaitsevo (Figs. 1, 2). We collected 7 samples (14 specimens) of grey bed horizons from 3 levels and 51 samples (72 specimens) of red beds from 28 levels from the Kartamysh Formation, and 6 samples (7 specimens) of red beds from 3 levels from the Nikitovskaya Formation. Some of the data from the Permian were reported in summary form by Iosifidi and Khramov (2002) — they are reported in full here, with new additional analyses. The Araucarite and Avilov Formations comprise red-brown, brown-red, brown, and gray-green, often sandy, clays alternating with sandstones of greenish-gray

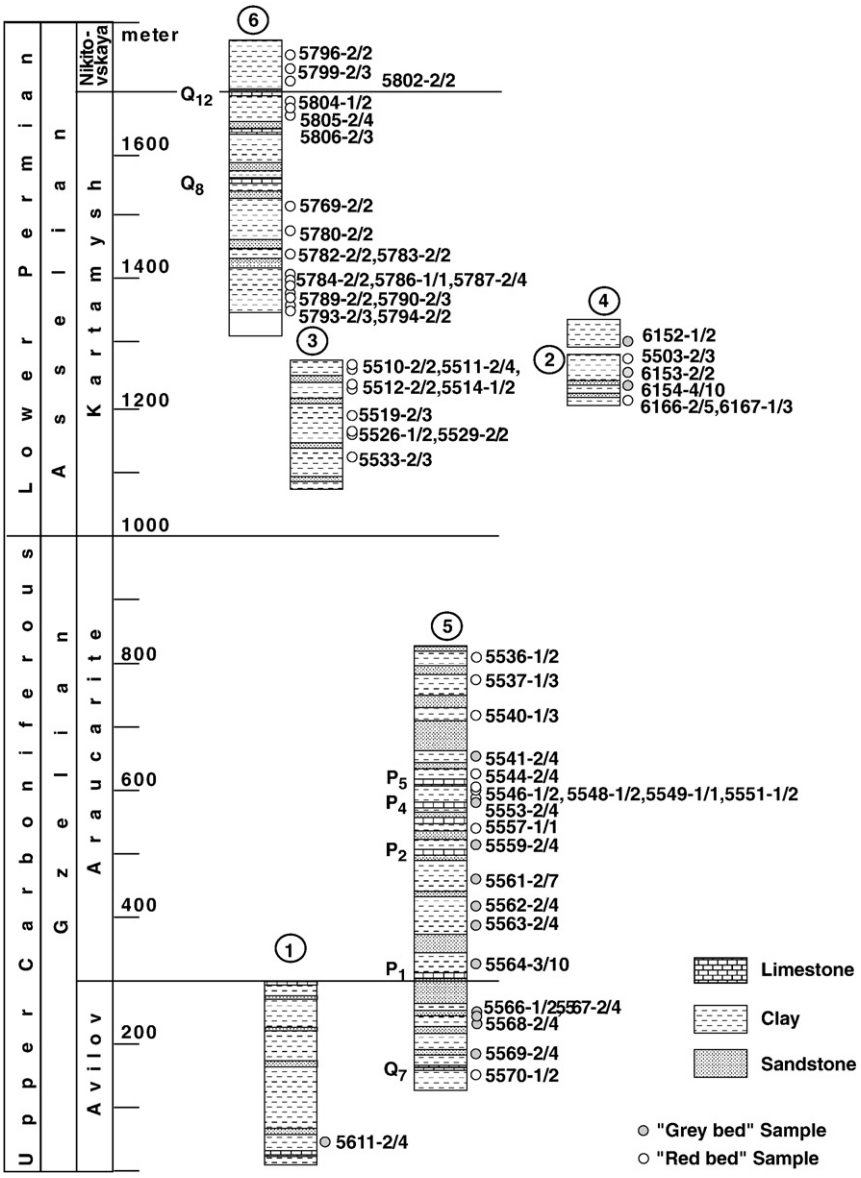


Fig. 2. Composite schematic lithological sections of the sampled localities. The investigated levels are indicated with open (red bed lithology) and shaded (grey bed lithology) circles. The four-digit number refers to the section level labelling, the other numbers indicate the number of block samples and the total number of specimens processed. O_7, P_{1-2} and Q_8, Q_{12} refers to the index of fossiliferous limestone guide beds. For section numbering see the caption to Fig. 1.

and yellowish-red colour, occasionally pink and red-brown. The conformably overlying Kartamysh and Nikitovskaya Formations (Asselian stage) comprise mostly red clays and cupriferous sandstones. Samples come from different fold limbs formed during the afore-mentioned Late Paleozoic orogenic event. The dip azimuths of the bedding at the sampled localities (Fig. 1) vary from 170–220° to 340–20°, their dip angles from 5° to 88°, enabling the application of a fold test.

4. Palaeomagnetic analysis

Measurements of the NRM were carried out using 2G 3-axis cryogenic magnetometers at the Universities of Munich and Utrecht, or using JR-4 and JR-5 spinner magnetometers in St. Petersburg. After thermal demagnetization at 100 °C (in most samples equivalent to the removal of a component related to laboratory storage), the NRM intensities typically vary between 0.8 and 10 mA/m in the red beds and between 0.1 and 5 mA/m in the grey beds, which is two-to-four orders of magnitude above instrumental noise levels. The anisotropy of magnetic susceptibility (AMS) was measured using a KLY-3 kappabridge in the St. Petersburg laboratory (AGICO, Brno, Czech Republic). Typical values of bulk susceptibility range from ~100 to ~250×10^{-6} SI units. Specimens were then subjected to progressive thermal demagnetization up to 700 °C, using at least 12 steps.

Demagnetization results were visually inspected in orthogonal and stereographic projections; characteristic remanence components were identified using least-squares algorithms (Kirschvink, 1980; Enkin, 1994). Components were considered stable where they were defined by at least 3 consecutive points on orthogonal projections and had a maximum angular deviation not exceeding 10°. Where multiple specimens were obtained from the same hand samples the directions of the individual specimens were averaged to produce a sample mean (see Table 1).

5. Palaeomagnetic results

All results are compiled in Table 1 and summarised in Figs. 3–5. Below we describe the main outcome grouped by age and lithology.

Table 1
New palaeomagnetic directions and poles from Upper Carboniferous and Lower Permian Donets Basin rocks (48.5°N; 37.8°E). *N*/*n* refers to number of samples and specimens; C.S. is the coordinate system: g = geographic, s = stratigraphic ; *D* and *I* are the declination and inclination (both in degrees) of the mean directions; *K* is the concentration parameter; α_{95} is the radius of the confidence circle at 95% probability for the average direction; Under 'Test' F_+, *F*–, *F*~ indicates that the fold test according to Watson and Enkin (1993) is positive, negative, indeterminate; Pole lat and Pole long are the Northern latitude and Eastern longitude of the paleomagnetic pole; A_{95} = semi-axis of circle of 95% confidence about the mean pole; the statistics are at sample level (for component *A* at specimen level). Plat. is the paleolatitude. Palaeomagnetic directions and poles from Upper Carboniferous and Lower Permian Donets Basin rocks (48.5 °N; 37.8 °E).

Number	Object of study, Donets Basin	Rock age	NRM component	Identification intervals, °C	*N*/*n*	C. S.	*D*°	*I*°	*K*	α_{95}	Test	Pole lat (°N)	Pole long (°E)	A_{95}	Plat(°)
1	Redbeds (clay)	C_3gz	B_r	300–650,675	8/15	g	216	36	17	14	F_+	42	176	4.6	8
						s	210	−15	108	5					
			C_r	650–675	7/10	g	223	32	4	35					
						s	213	−5	86	7	F_+	36	175	6.3	2
2	"Grey" (clay, sandstone)	C_3gz	A_g	20–250	20/39	g	347	66	14	6	F_-	81	313		49
			B_g	250–430,575	16/29	g	219	51	13	11	F_+	41	181	4.6	6
						s	207	−11	58	5					
3	Mean ($B_r + B_g$)	C_3gz	B_{rg}		24/44	g	218	46	14	8					
						s	208	−13	67	4	F_+	42	179	3.3	6
4	Redbeds (clay)	P_1as	B_r	300–650,675	28/51	g	235	−38	19	6					
						s	217	−25	108	3	F_+	43	164	2.3	13
			C_r	650–675	21/36	g	230	−17	17	8					
						s	224	−2	53	4	F_+	30	165	3.9	1
5	"Grey" (clay)	P_1as	A_g	20–250	6/13	g	346	74	30	8	F_-	76	7		60
			B_g	400–680	6/13	g	223	−8	97	7					
						s	221	−10	102	7	F~	34	165	5.7	5

5.1. Upper Carboniferous

5.1.1. Red beds

Thermal demagnetization of the red mudstones and clays of the Avilov and Araucarite Formations yielded demagnetization diagrams that are straightforward to interpret. Most samples yield evidence for two or three NRM components: a low temperature component "A_r" (T_{ub}<250 °C) and two high temperature components labelled "B_r" (300 °C<T_{ub}<650–675 °C) and "C_r" (T_{ub}>650 °C; see Fig. 3A and B). The unblocking temperatures of the "B_r" and "C_r" components are

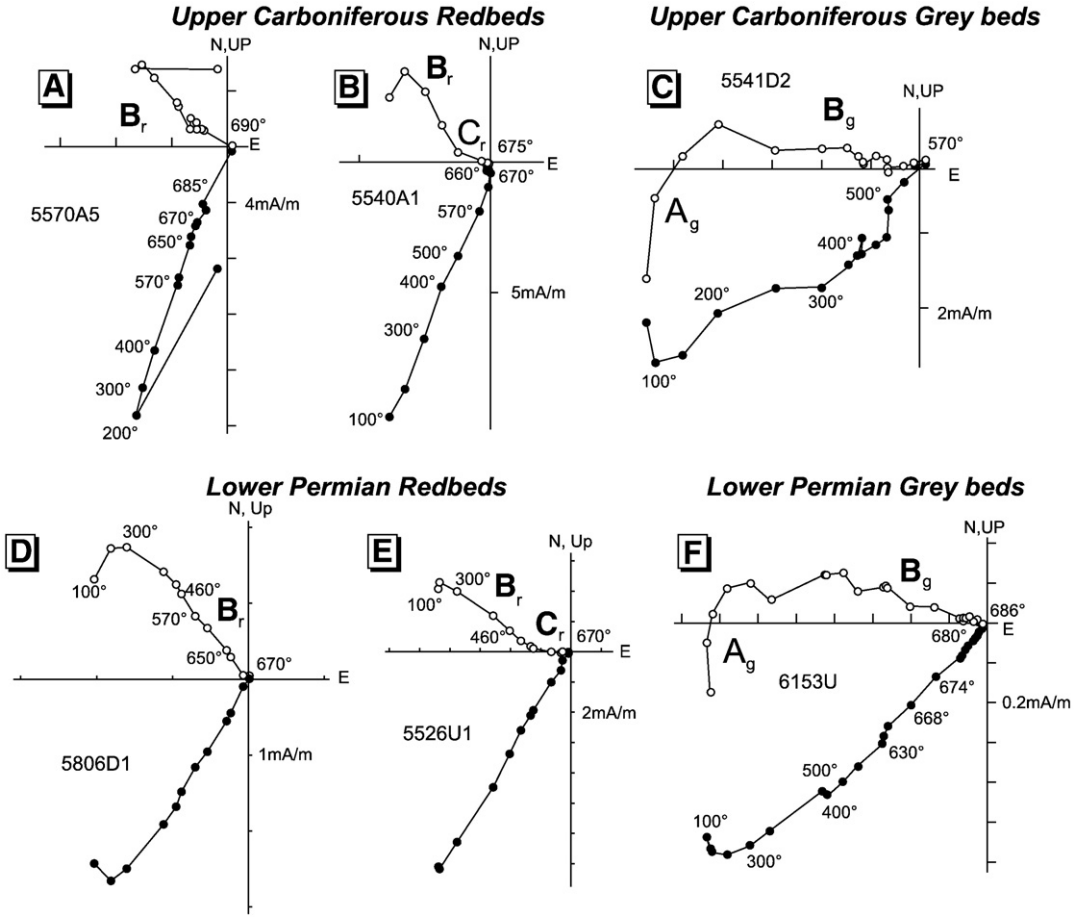


Fig. 3. Zijderveld plots (stratigraphic coordinates) of typical thermal demagnetization of the Upper Carboniferous (A–C) and Lower Permian (D–F) rocks from the Donets Basin sections. Closed (open) circles indicate the projections onto the horizontal (vertical) plane. Temperatures are degrees Centigrade. When distinguishable the C components are always shallower than B components. For full explanation see text.

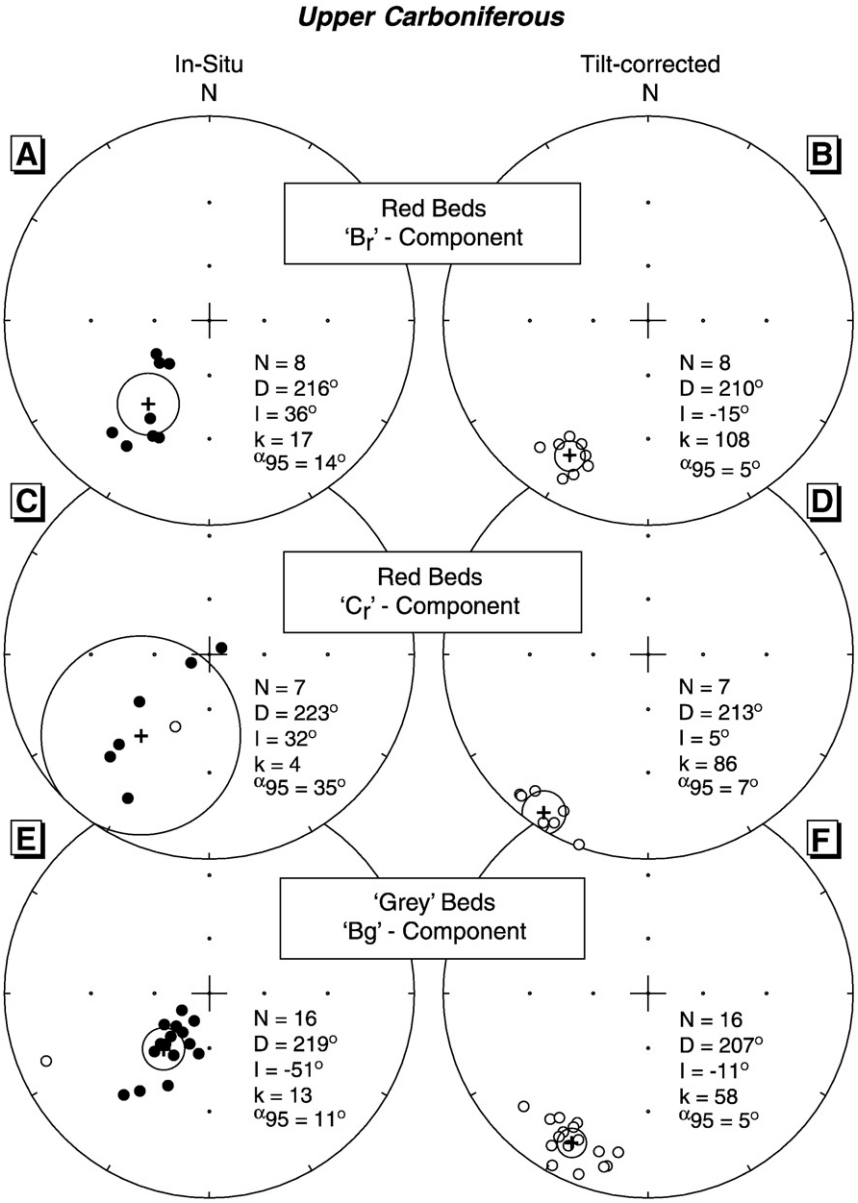


Fig. 4. Equal area plots of directions of the identified NRM components for the Upper Carboniferous rocks. Closed (open) symbols are in the lower (upper) hemisphere. Average directions and 95% confidence circles are indicated. Tilt-correction (right-handed panel column) improves grouping pointing toward prefolding components; full explanation in text.

consistent with being carried by haematite. The low temperature component "A_r" was typically found in all samples, but the direction of this component varied from sample to sample (cf. Fig. 3A, B, D and E). Given that the samples had been stored in a laboratory field for almost 40 years we suspect that this "A_r" component is most likely a recent viscous remanent magnetization (VRM) acquired during storage. The higher temperature component "B_r" was recovered from eight samples (15 specimens). The "B_r" component is generally directed with shallow to intermediate downward inclinations to the south in geographic coordinates (Fig. 4A). Tilt correction brings a pronounced improvement in the grouping of the directions (Fig. 4B), and the data pass the fold test of Watson and Enkin (1993) at the 95% confidence level, indicating that the "B_r" component predates folding.

The second high temperature component "C_r" was identified in seven samples (10 specimens). The "C_r" component is somewhat scattered in geographic coordinates (Fig. 4C) and the grouping improves substantially on tilt-correction (Fig. 4D), again indicating that the "C_r" predates folding. It is noteworthy that the major difference between the "B_r" and "C_r" components in their stratigraphic (i.e. tilt-corrected) coordinates is in inclination, with the "C_r" component being consistently shallower than the "B_r" component.

5.1.2. Grey coloured rocks

Thermal demagnetization of the grey sandstones and clays from the Avilov and Araucarite Formations generally yielded evidence for two components of magnetization (Fig. 3C): a low temperature (T_{ub}<250 °C) component "A_g" and a high temperature (300–500 °C<T_{ub} <570 °C) component "B_g". The low temperature component "A_g" was recovered from twenty samples (39 specimens) and, in contrast to the "A_r" component in the red beds, is well grouped in geographic coordinates in a direction that is close to the present-day Earth's field (PEF) at the sampling site. Tilt-correction of the "A_g" component directions brings an increase in the dispersion and the data fail the fold test at 95% probability. Therefore, this "A_g" component is most simply a VRM of recent age, acquired during the Brunhes Chron. The distribution of directions for the "B_g" component is similar to that of the "B_r" component in the red beds (cf. Fig. 4A and E), and tilt correction brings an improvement in the grouping (Fig. 4F), yielding a positive fold test at the 95% confidence

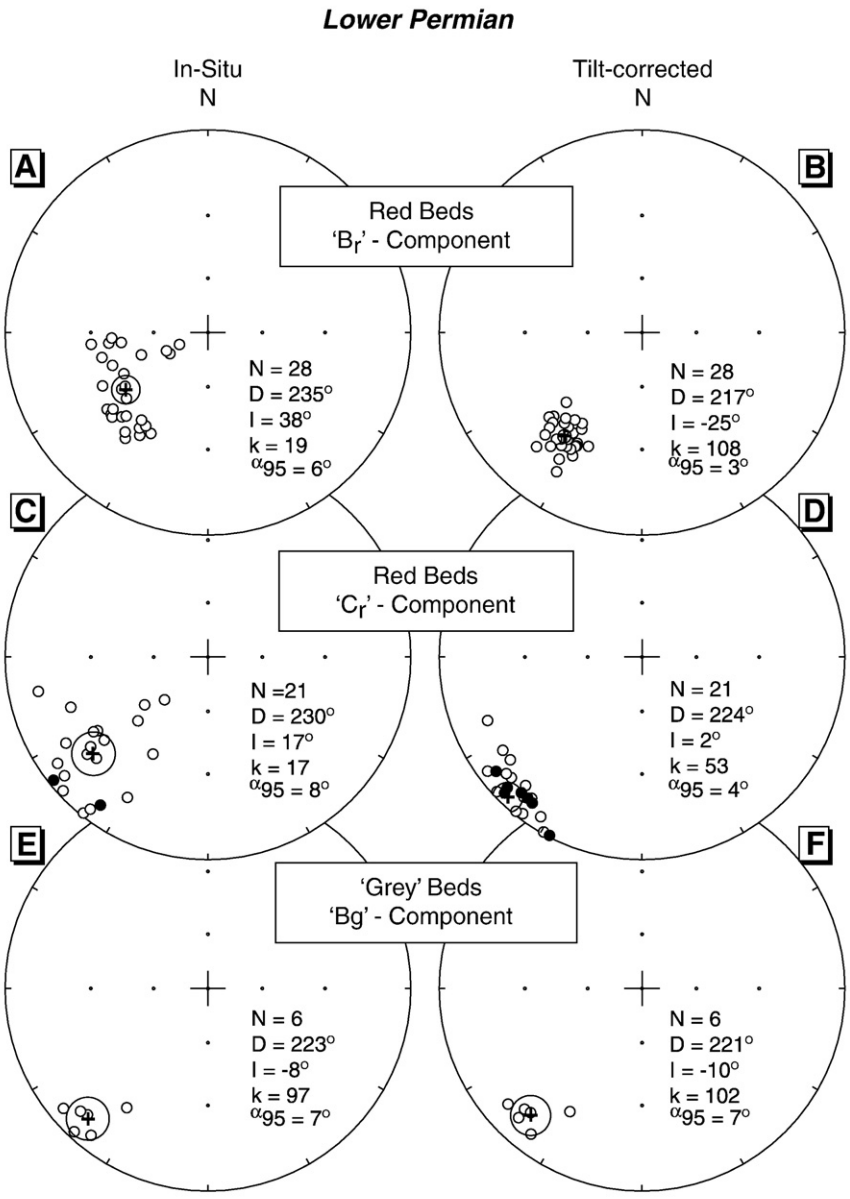


Fig. 5. Equal area plots of directions of the identified NRM components for the Lower Permian rocks. Closed (open) symbols are in the lower (upper) hemisphere. Average directions and 95% confidence circles are indicated. Tilt-correction (right-handed panel column) improves grouping (note that the B_g component barely changes) pointing toward prefolding components; full explanation in text.

level. This indicates that the "B_g" component in the grey sediments also predates folding. An important difference to the red beds, however, is that the unblocking temperatures of the "B_g" component in the grey beds are consistent with magnetite (or low-Ti titanomagnetite) being the principal remanence carrier.

5.2. Lower Permian

5.2.1. Red beds

As with the Upper Carboniferous units, most Lower Permian red bed samples yield evidence for two or three components of magnetization: a low temperature (T_{ub}<250 °C) component "A_r" and two high temperature components "B_r" (300 °C<T_{ub}<650–675 °C) and "C_r" (T_{ub}>650 °C see Fig. 3D and E). The directions of the low temperature "A_r" component are scattered from sample to sample, and we again interpret them to be of viscous origin, acquired during storage.

The unblocking temperatures of the "B_r" and "C_r" components are again consistent with haematite being the predominant remanence carrier. The "B_r" component was recovered from 28 samples (51 specimens) in the red beds, and is generally directed with shallow to intermediate upward pointing inclinations, to the south-west in geographic coordinates (Fig. 5A). Tilt-correction brings about a notable improvement in the grouping of the mean directions (Fig. 5B), and the data pass the fold test of Watson and Enkin (1993) at the 95% confidence level, indicating that the "B_r" component predates folding.

The second high temperature component "C_r" was identified in twenty one samples (36 specimens). The "C_r" component is rather scattered in geographic coordinates (Fig. 5C) and the grouping improves substantially on tilt-correction (Fig. 5D), again indicating that the "C_r" component predates folding. Similarly to the Upper Carboniferous red beds, the major difference between the "B_r" and "C_r" components in the Permian red beds is that the "C_r" component is

consistently shallower than the "B_r" component in stratigraphic coordinates.

5.2.2. Grey coloured rocks

Thermal demagnetization of the grey clays from these formations revealed two NRM components (Fig. 3F): a low temperature component "A_g" (T_{ub} <250 °C); and a high temperature component "B_g" (300 °C< T_{ub} <680 °C). The "A_g" component is close to the PEF in geographic coordinates, fails a fold test, and is likely a VRM of recent age. The "B_g" component was identified in six samples (13 specimens) and is generally directed with shallow upward inclinations to the southwest (Fig. 5E). Tilt correction brings a very slight improvement in its grouping; however the improvement is statistically insignificant (Fig. 5F). It should be noted, however, that the unblocking temperatures for the "B_g" component in the Permian grey beds are consistent with the major remanence carrier being haematite, in spite of the greyish colour of the sediments. It is also noteworthy the mean direction of the "B_g" component in the grey beds is intermediate between that of the "B_r" and "C_r" components in the coeval red beds.

6. Anisotropy of magnetic susceptibility

To try to see whether the recording of the "B" or "C" components might be influenced by any fabrics present within the rocks we examined the anisotropy of magnetic susceptibility (AMS) for the Permian units (unfortunately demagnetization experiments had been carried out on the Upper Carboniferous units before AMS measurements could be made). The results are presented in Fig. 6, which illustrates the AMS fabrics measured for the grey beds (Fig. 6A); for some of the red beds that yield only the "B" component (Fig. 6B); and for some of the red beds that also yield the "C" component (Fig. 6C). All samples appear to show a pronounced bedding-parallel foliation, with in some cases a very high intensity. Lineations are up to a few percent while foliation ranges up to 20%. There does not, however, seem to be any direct correlation between the intensity of the fabric and the presence of either a "B" or "C" component in the samples (Fig. 6D). It may be that the AMS fabric is not directly related to the mineralogy of the remanence carriers. It is plausible that there is a considerable paramagnetic contribution to the low-field susceptibility of the samples, particularly for the clay-rich samples, which may dominate or swamp the ferromagnetic contribution to the susceptibility. Alternately it might be that the AMS fabric is dominated by the finest haematite grains, and the remanence is dominated by a coarser haematite fraction (e.g. Tan et al., 2003). This would suggest that measurements of AMS are of limited value in assessing whether or not inclination shallowing has occurred within the sedimentary package, unless the ferromagnetic contribution is high; a conclusion similar to that reached by Tan et al. (2002, 2003) and Kodama and Dekkers (2004).

7. Discussion

The mean palaeomagnetic directions from all studied units are plotted in Fig. 7. In both the Upper Carboniferous and Lower Permian

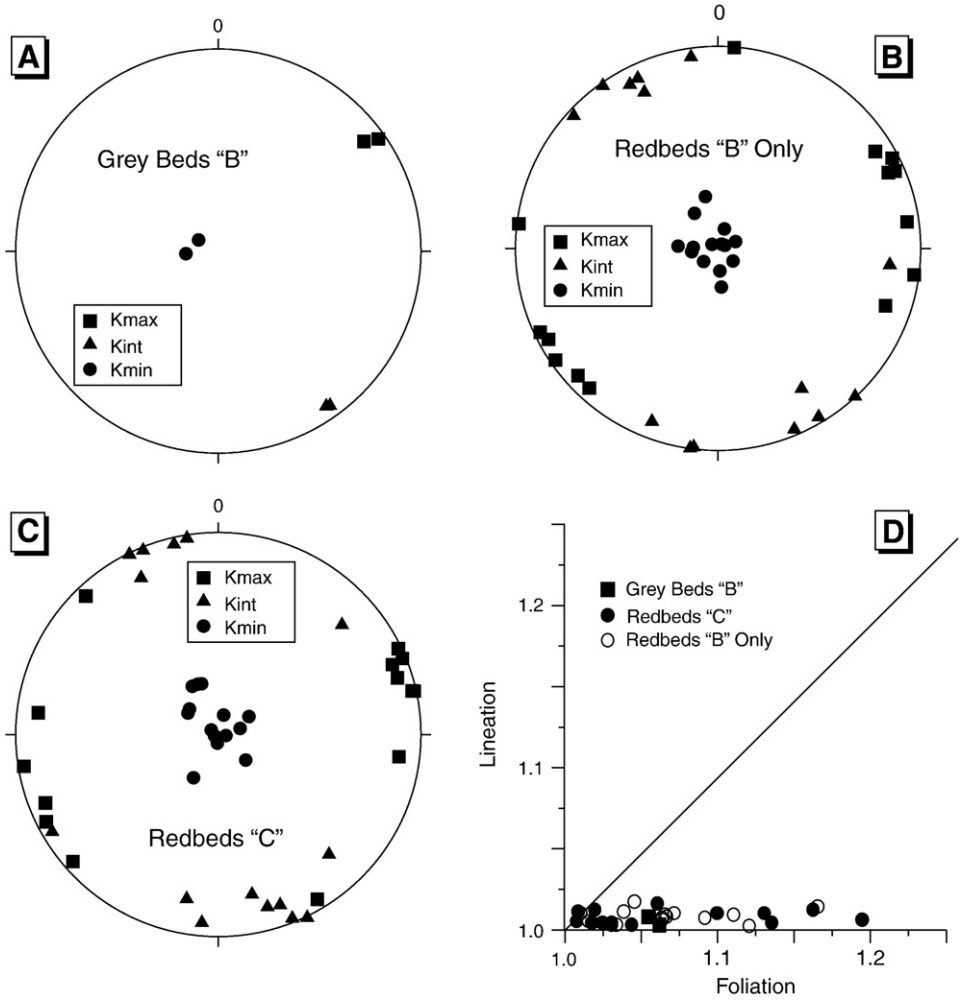


Fig. 6. AMS behaviour of selected samples of the Lower Permian sediments. Equal area plots of principle axes of A) grey beds, B) red beds with B component only, C) red beds with components B and C occurring simultaneously. D) Flinn diagram showing dominant folation (up to almost 20%) and low lineation only (up to ~2%). There appears to be no obvious relation between AMS properties and lithology.

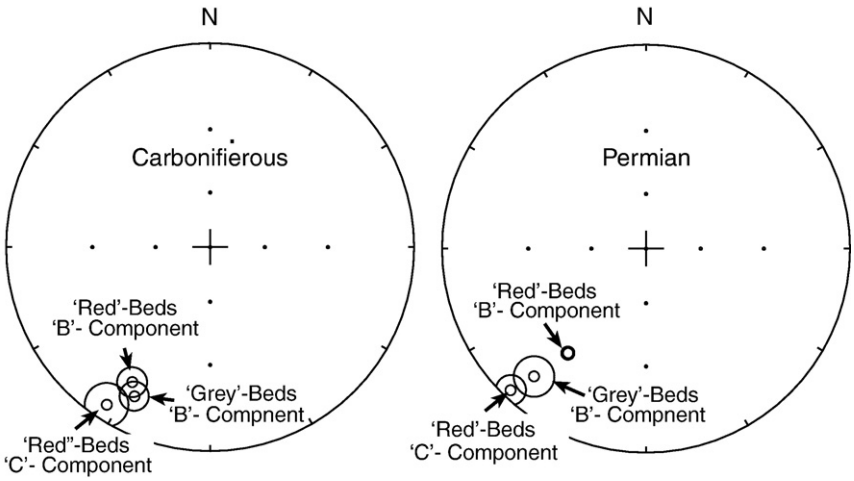


Fig. 7. Summary plots of the directions for the NRM components retrieved from the Upper Carboniferous (left panel) and Lower Permian (right panel) sediments. Note that the Carboniferous grey bed B component is shallower than the coeval red bed B component. The red bed C component is the shallowest In the Permian sediments the same situation occurs. The red bed B components are residing in pigmentary haematite and have formed during an early reddening phase. Therefore they are least influenced by inclination shallowing. Further explanation in text.

red beds the "C_r" component directions are consistently shallower than the corresponding "B_r" component directions, and both components pass fold tests indicating that they predate folding. The deformation in this area started in the Upper Carboniferous but main phase of folding was from post-Asselian to terminal Lower Permian time (Kirikov, 2006).

In the case of the Upper Carboniferous rocks, the direction of the "B_g" component in the grey beds is similar to that of the "B_r" in the red beds, whereas for the Lower Permian rocks the direction of the "B_g" component in the "grey" beds is closer to that of the "C_r" component in the red beds. A method for correcting such data for 'inclination-shallowing' has recently been proposed (e.g. Tauxe and Kent, 2004; Tauxe et al., 2008) – the so-called elongation/inclination (E/I) method – which compares the distribution of palaeomagnetic inclinations in a study to a statistical model for the paleosecular variation of the field. The method allows for the directions at a site, or in a group of sites, to be progressively 'unflattened' until the distribution of inclinations matches that predicted by the field model. For example Meijers et al in press use this approach in their analysis of the Permo-Carboniferous strata of the Donbas fold belt, because their study had a sufficient number of samples to do so. Our sample numbers are rather limited, which precludes a meaningful application of the E/I method, but it should also be noted that Tauxe et al. (2008) argue only that the use of the method is warranted for Mesozoic and Cenozoic times — caution should exercised in its application to Palaeozoic rocks because the palaeosecular variation model that underlies the method has not yet been thoroughly tested for Palaeozoic times.

One major difference between the Upper Carboniferous and Lower Permian grey beds is that the dominant magnetic carrier in the older rocks is magnetite, whereas it is haematite in the younger rocks. Thus it seems likely that the major cause of the directional discrepancy is related to the nature of the remanence carrier.

Petrographic analysis of the studied units provide some further constraints on the relation between the presence of particular components and the magnetic mineralogy. Samples containing the very shallow "C_r" component in both Upper Carboniferous and Lower Permian rocks were found to contain large (0.01–0.10 mm) flat particles of haematite (Fig. 8). In contrast, samples carrying only the "B_r" component were characterized by the presence of pigmentary haematite, and an absence of detrital grains. This is further confirmed by the presence of large detrital haematite grains in the Permian "grey" beds, which may indicate why they have a shallower "B_g" component than the coeval red beds, and unblocking temperatures that are consistent with a haematite carrier for the NRM. All samples containing large detrital haematite have undergone inclination shallowing as a result of the flaky nature of specular haematite and the depositional process that is biased towards the horizontal. The samples containing dominantly pigmentary haematite have undergone less shallowing that could be amenable to correction with the E/I method. A likely sequence of events, therefore, is that the C component was acquired as a DRM during deposition and was then subsequently mechanically rotated towards the horizontal during compaction and lithification of the host rocks. In the case of the grey beds the B component is carried by magnetite in the upper Carboniferous beds and by haematite in the lower Permian beds. The magnetite-hosted "*B*" component may have undergone a small degree of shallowing in the case of the Upper Carboniferous grey units (Fig. 7), whereas the haematite-hosted "B" component has undergone a significant degree of shallowing in the case of the Lower Permian grey beds. This is consistent with the morphology of the respective magnetic carriers, as magnetite tends to be equant or elongate, whereas specular haematite is more platy, and, hence, more prone to mechanical reorientation. The pigmentary haematite likely developed during diagenesis and acquired a CRM which records a high-fidelity "*B*" component. This CRM must have been acquired relatively early in the history of the rock as it passes the fold-test.

In calculating our mean Upper Carboniferous palaeopole for the "B" component we have combined the "B" directions from both the red beds and grey units, as they pass the common true mean direction (ctmd) test which is equivalent to the reversal test of McFadden &

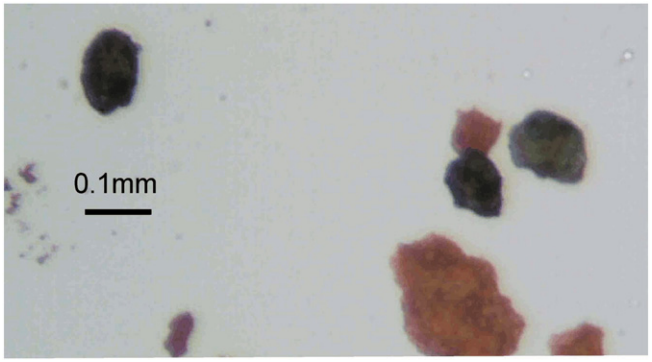


Fig. 8. Magnetic extracts from Sample 5805 (Lower Permian) illustrating the presence of large 0.01–0.10 mm) flat particles of haematite in samples containing the shallow "C" component of magnetization.

McElhinny (1990) with a classification B ($\gamma = 5.0 < \gamma_c = 7.0$). The "*B*" component directions in the Lower Permian red and grey rocks do not pass the common true mean direction test, and the poles are, therefore, calculated separately.

A comparison of the resulting palaeopoles with reference palaeomagnetic data for the East European platform (Iosifidi and Khramov, 2002) and with an apparent polar wander path from the data compilation of Torsvik et al. (2008) is presented in Fig. 9. To calculate the path we have taken the European reference data of Torsvik et al. (2008) and applied a spherical spline with a smoothing factor of 200 (e.g. Torsvik and Smethurst, 1999). The "B" component directions from the rocks with either a magnetite or pigmentary haematite carrier are in general agreement with the reference data, whereas the "B" and "C" component directions carried by detrital haematite are consistently shallower that the reference directions. The "C" component in the Carboniferous red beds does overlap the reference pole, though this is likely due to the fact that the study location was closer to the equator in the late Carboniferous, and shallowing has less effect on inclinations that are already close to horizontal.

The resulting palaeolatitudes are presented in Fig. 10 (see also Table 1) along the palaeolatitudes calculated for the study area from the European reference dataset of Torsvik et al. 2008. For our upper Carboniferous data there is a spread of 6° of latitude between the "B" and "C" components of magnetization in the redbeds and 4° of latitude between our overall "B" component mean (from both the red and grey lithologies) and the "C" component. This would translate into a ~660 or ~440 km difference in the palaeolatitude. For our lower Permian data there is a 12° latitudinal difference between the palaeogeographic position indicated by the "B" component and "C" components in the redbeds. This translates into a ~1330 km difference in the

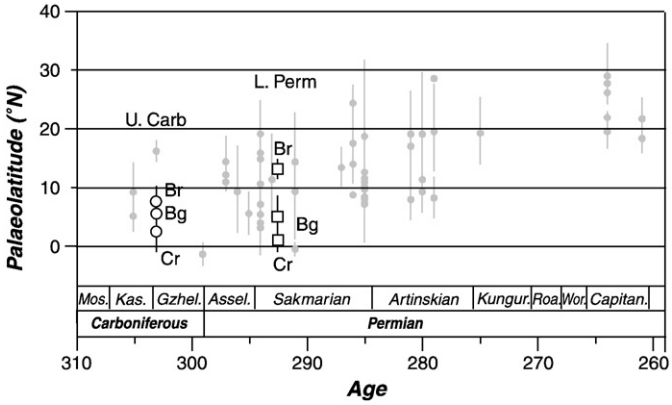


Fig. 10. Palaeolatitudes, with associated errors, calculated for the study area (48.5°N; 37.8°E) plotted against time. Open circles are our upper Carboniferous data, and open squares are our lower Permian data. The labels against the symbols are the different components of magnetization identified in our study (see text for details). The grey closed circles, and associated errors, are the palaeolatitudes calculated for our study from the European reference data compilation of Torsvik et al. (2008). Abbreviations for the stage names are: Mos. = Moscovian; Kas. = Kasimovian; Gzh. = Gzhelian; Assel. = Asselian; Kungur. = Kungurian; Roa. = Roadian; Wor. = Wordian; Capitan. = Capitanian.

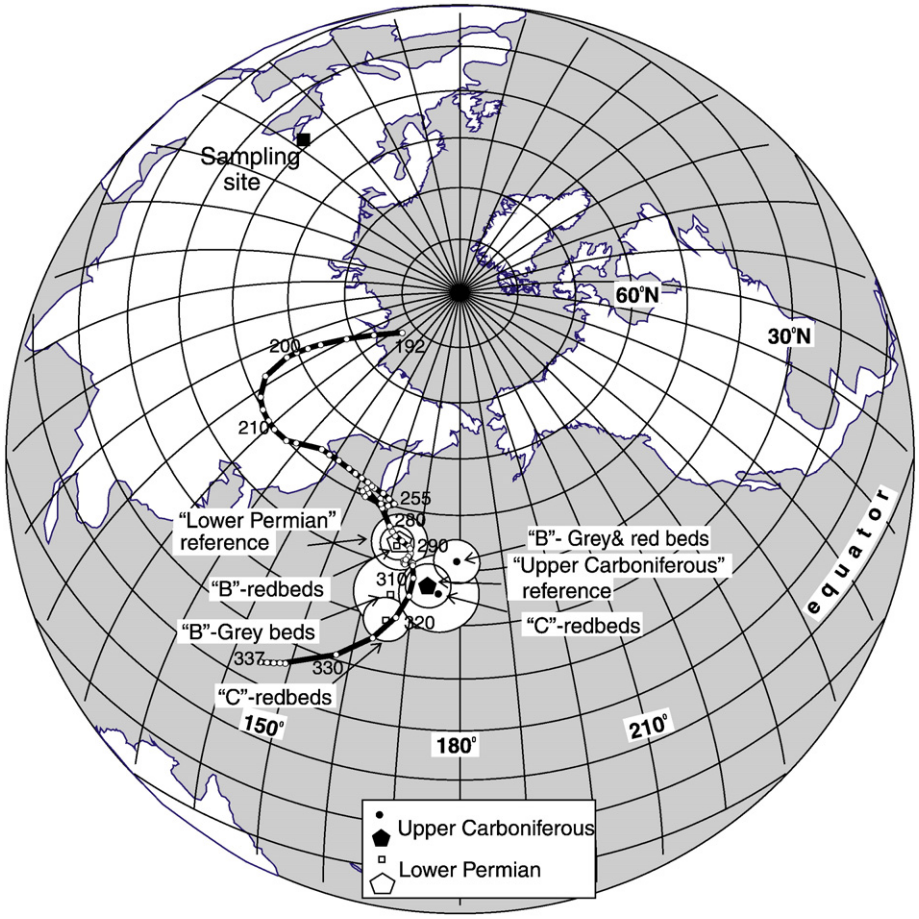


Fig. 9. Pole positions for the Upper Carboniferous and Lower Permian rocks from Donets Basin, with associated A_{95} values. The East European Platform Upper Carboniferous, and Lower Permian reference poles (Lat = 37°N, Long = 173°E, $A_{95} = 4°$; and, Lat = 44°N, Long = 165°E, $A_{95} = 5°$ respectively) of Iosifidi and Khramov, 2002 are also plotted as hexagons. Poles for the Upper Carboniferous sediments are indicated with open circles (red and grey beds) while poles for the Lower Permian sediments (red and grey beds) are indicated with open squares. The APW path was generated from the European data compiled by Torsvik et al., 2008, with a spherical spline (smoothing factor 200) fitted to the data. Numbers adjacent to the path are the ages (in Ma) assigned by the spline.

palaeolocation of the study area. Given that both components pass fold tests it is only because we were able to link the components to specific remanence carriers that we were able to deduce the order of magnetization. It is also clear from Fig. 10 that this spread of palaeolatitudes is not restricted to our study. We calculated differences of more than 17° of latitude for Artinskian time, depending on the reference data chosen, so it is clear that at least some of the reference datasets also contain data where pronounced degrees of inclination shallowing have gone undetected in the original studies. Such data may also be key in addressing palaeogeographic controversies at this time, such as the various proposed Pangea fits. Although a complete discussion of the relative merits of the different proposed Pangea fits is outside the scope of the present paper, we argue that the solution to such controversies will likely come from a more selective approach to choosing reference data, rather than following the recent trend of interrogating large databases in the hope that the signal outweighs the noise.

8. Conclusions

Upper Carboniferous to Lower Permian lithologies of the Donets Basin carry two ancient components of magnetization: a "B"-component with a wide (300–670 °C) range of unblocking temperatures, and a "C" component with a narrow range of unblocking temperatures. Both components pass fold tests indicating that they were acquired early during the history of the rocks. The "B" component is carried by magnetite and/or fine-grained pigmentary haematite, whereas the "C" component is carried by large specular haematite grains. The "C" component has undergone a significant degree of inclination-shallowing, which produces up to 12° of latitudinal mismatch in reconstructions compared with "B" component. We argue that European reference palaeomagnetic data are also contaminated by undetected cases of inclination-shallowing in the original studies, and a more robust approach to data selection may be required to address key palaeogeographic controversies for Permo-Carboniferous time.

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